Topic 3 Test

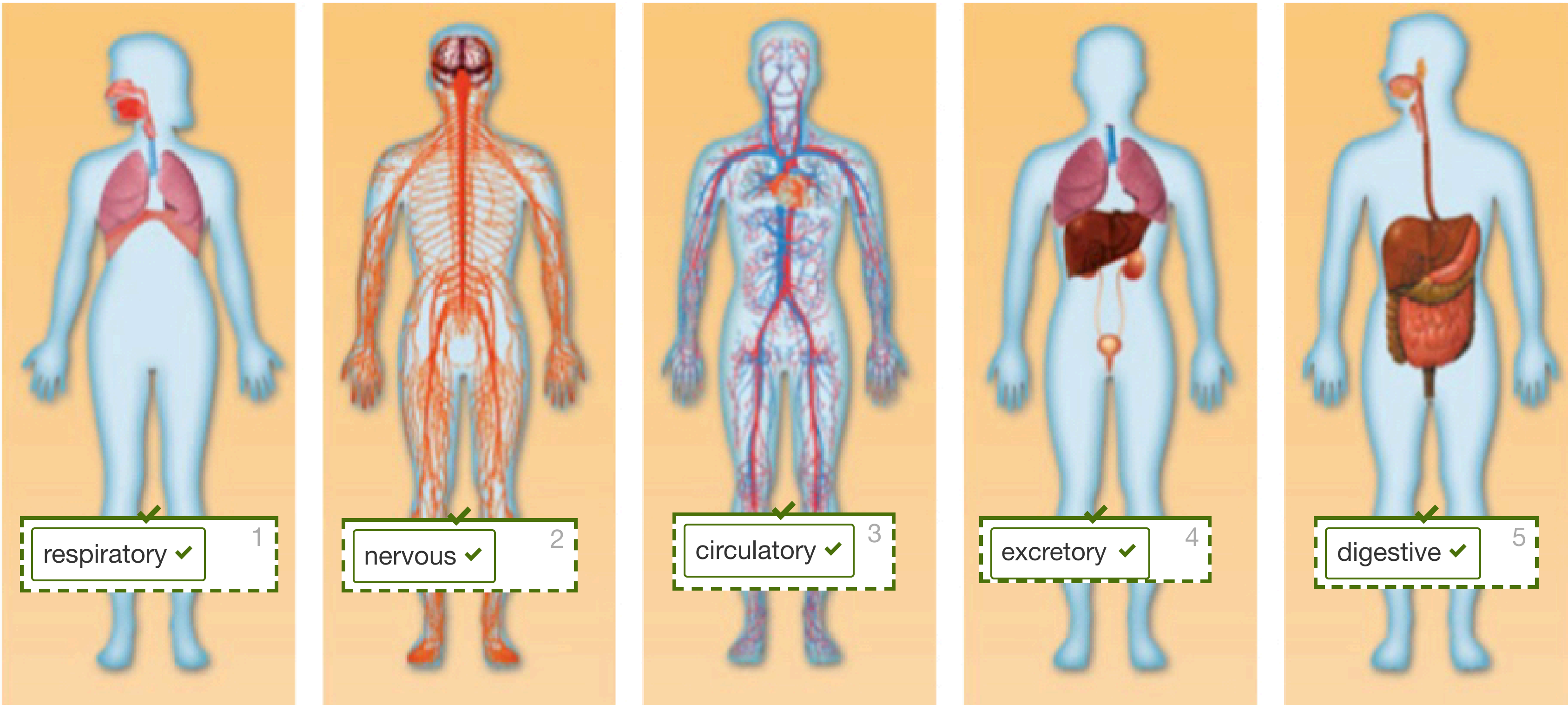
ATTEMPT SCORE

22.666667 / 40

01 Label Image **5** / 5

Matching.

Match the organ system to the correct diagram. Not all system names will be used!



immune muscular skeletal

Given responses:

1 respiratory

2 nervous

3 circulatory ✔

4 excretory ✔

5 digestive ✔

02 Multiple Choice

2 / 2

Choose the best answer.

When internal body temperature drops below 98.6˚, skeletal muscles may begin to twitch to create heat. Shivering, then, is an example of the body maintaining what?

- cellular respiration
- excretion
- homeostasis ✔
- peristalsis

03 Multiple Choice

0.67 / 2

Select all that apply.

Which nutrients can the body use for energy?

- water ✗
- proteins ✓
- vitamins
- carbohydrates ✓
- fats ✓
- minerals

04 True/False – / 2

True or false.

Digestion happens before absorption.

- True ✓
- False

05 Multiple Choice 2 / 2

Choose the best answer.

What eventually happens to fluid that leaks from capillaries into the surrounding tissues?

- it replaces dead red blood cells
- it returns to the blood through the lymphatic system ✔
- it gets absorbed by the large intestine
- it gets excreted through the skin

06 Multiple Choice 0 / 2

Choose the best answer.

A red blood cell that just came from your right big toe arrives in your heart, where does it go next?

- back to the right big toe
- brain
- lungs ✔
- it could go anywhere! ✘

07 Multiple Choice 2 / 2

Choose the best answer.

What is the name of the process that uses oxygen to turn glucose into usable energy for cells?

- peristalsis
- cellular respiration ✔
- excretion
- chemical digestion

08 Short Answer/Essay

– / 3

Short answer.

Why is the small intestine lined with villi? And why are the lungs full of alveoli? What do these structures (villi and alveoli) have in common, and how do they help their respective organs perform their functions?

The small in testine is lined with villi because of a specific reason. See, the human body needs to absorb as much nutrients as it can to help your body maintain homeostasis. So the small intestine uses villi to reach into the food to grab nutrients that are needed such as proteins, carbs, vitamins, and fats. The lungs also have something similar to villi called alveoli. Alveoli are structures that help supplement the circulatory system with oxegen. In a picture of an alveoli, there are many blood vessels that surround the alveoli. This is so the oxygen can get into the bloodstream. Alveoli and villi are two structures that have similiar uses. For example Alveoli and villi are used to absorb as many nutrients as possible. Many alveoli are in the lungs to help the oxygen get into the bloodstream. Similar to that there are many villi in the small intestine to absorb as much oxygen as possible. The main job of the respiratory system is to get oxygen into the bloodstream. Allthe other organs in the system only work together to make the alveoli get the oxygen. The alveoli are the real deal, they are the main reason you have oxygen in your blood and the main reason carbon dioxide isn't stuffed inside your lungs. The villi are similar to this. Villi are in thousands in the small intestine so they absorb all the small nutrients the big intestine doesn't absorb. We would be dead without villi.

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09 Multiple Choice 2 / 2

Choose the best answer.

When a blood vessel breaks, platelets fuse to the broken blood vessel walls and release a chemical that attracts more platelets to the site. More platelets release more of the chemical that attracts more platelets.

What type of feedback loop is this?

- positive ✔
- negative

10 True/False – / 2

True or false.

The function of the kidneys is to filter waste materials, like urea, out of the blood.

- True ✔
- False

11 Multiple Choice 2 / 2

Choose the best answer.

Ron picks up a metal pan lid that was covering a pot of boiling water, and he drops it immediately. Cells in his fingers sent the message that his fingers had touched heat. This response is automatic and occurs without a conscious thought. What is this response an example of?

- neuron
- reflex ✔
- instinct
- synapse

12 Ordering 5 / 5

Ordering.

Scott and Stiles take a bike ride in the Beacon Hills Reserve. Both riders are careful to make space for other travelers on the path. Suddenly, Stiles sees a young deer getting ready to cross the trail right in front of them. “Watch out! Deer!” he yells. Scott has less than a second to react and avoid a crash with the deer. Number the steps below to put Scott’s reaction in the correct order (TOP TO BOTTOM).

- Scott’s ears detect Stiles’s shouted warning. ✔
- Scott’s brain directs his eyes to look for the deer. ✔
- Scott’s eyes focus on the deer. ✔
- Scott’s brain interprets the images from his eyes and sends commands to his muscles. ✔
- Scott’s muscles move the bike out of the way of the deer. ✔

13 Multiple Choice 2 / 2

Choose the best answer.

The brain and spinal cord make up what part of the nervous system?

- ○ autonomic
- ◉ central ✔
- ○ peripheral
- ○ somatic

14 True/False – / 2

True or false.

The nerves responsible for controlling smooth muscle in the stomach are part of the somatic nervous system.

- True ✗
- False ✔

15 Short Answer/Essay – / 5

Short answer.

Describe how different organ systems work together to provide each and every cell in the body with glucose and oxygen. (*Use a minimum of three systems in your answer and name at least one organ per system. Maximum 250 words.*)

Different organ systems work together to help the human body. Lets consider how three systems work together to provide every cell with glucose and oxygen. First your respiratory system helps you bring in oxygen through the bronchi, into the lungs, and into the alveoli. Next, the alveoli release the carbon dioxide back into the lungs and deposits the oxygen into the circulatory system. From there it travels to the heart and goes on its journey through the body in ateries. Meanwhile the digestive system's liver releases glucose into the bloodstream and into the body. Now the oxygen and the glucose travel in ateries to get the nutrients to every cell, from the tip of your head to the tip of your toe-nail. Capillaries are reach into the edge of every corner in your body to make sure that every cell gets its needed nutrients. The blood then circulates back into the heart through the veins. And the cycle repeats! Now this whole lot-a-stuff can't be done by a single system. So that's why it is important to have many systems. This is how the body gets to provide glucose and oxygen to every, miniscule cell of your body, fascinating.

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